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A Single-Loop Reliability-Based MDSDO Formulation for Combined Design and Control Optimization of Stochastic Dynamic Systems

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Optimization of dynamic engineering systems requires an integrated approach that accounts for the coupling between embodiment design and control system design, simultaneously. Generally known as combined design and control optimization (co-design), these methods offer superior system's performance and reduced costs. Despite the widespread use of co-design approaches in the literature, not much work has been done to address the issue of uncertainty in co-design problem formulations. This is problematic as all engineering models contain some level of uncertainty that might negatively affect the system's performance, if overlooked. While in our previous study we developed a robust co-design approach, a more rigorous evaluation of probabilistic constraints is required to obtain the targeted reliability levels for probabilistic constraints. Therefore, we propose and implement a novel stochastic co-design approach based on the principles of reliability-based design optimization (RBDO) to explicitly account for uncertainties from design decision variables and problem parameters. In particular, a reliability-based, multidisciplinary dynamic system design optimization (RB-MDSDO) formulation is developed using the sequential optimization and reliability assessment (SORA) algorithm, such that the analysis-type dynamic equality constraints are satisfied at the mean values of random variables, as well as their most probable points (MPPs). The proposed approach is then implemented for two case studies, and the results were benchmarked through Monte Carlo simulation (MCS) to indicate the impact of including reliability measures in co-design formulations.
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Keywords: design optimization, multidisciplinary design optimization, reliability in design, systems design, uncertainty analysis

1 Introduction

As dynamics—a system's state evolution through time—becomes an increasingly critical trait of engineering systems, it is essential to employ methods that account for the bi-directional dependency of embodiment design and control system design. In recent times, the optimality of conventional dynamic-system design solutions has been challenged by multiple novel approaches that take into account the coupling between plant design and control system design. Generally, the plant and controller problems can be solved using sequential, iterative, nested, and simultaneous approaches [1,2], where the nested and simultaneous strategies produce a system-level optimal solution. These latter two approaches, which are classified as combined design and control optimization (co-design) methods, have been extensively studied in the literature for deterministic dynamic systems.

The early work in the deterministic co-design demonstrated its benefits through comparisons with the conventional sequential and iterative approaches. A combined optimal design and control formulation of a DC electric motor was examined in Ref. [3], where the authors concluded that the separation of optimal design and optimal control does not lead to an optimal system solution. Fathy et al. demonstrated the superior performance of an automotive suspension system when formulated in a nested co-design approach [4]. Control proxy functions were employed by Peters et al. to obtain system optimal results with the simplicity of sequential optimization [5]. Later, Allison et al. solved a simultaneous co-design formulation for an active suspension system, using direct transcription (DT) [6]. They also concluded that the simultaneous co-design approach achieves better performance when compared with the sequential design. Likewise, Deshmukh and Allison improved the energy production of a wind turbine by 8% using simultaneous and nested co-design formulations [6]. More recently, Azad et al. [8,9] solved a simultaneous co-design problem of a moderate-fidelity plug-in hybrid electric vehicle powertrain. The authors achieved a system-level optimal solution by including performance criteria such as 0–60 mph acceleration time and all-electric range in the co-design formulation. Clearly, all of these studies indicate the effectiveness of co-design problem formulations in complex engineering design.

Despite the success of co-design for deterministic applications and with the exception of a new study that compares the simultaneous and nested formulations for the double-loop reliability-based co-design in Ref. [10], not much research has been performed on the co-design of stochastic dynamic systems. This is problematic as all engineering models contain some level of uncertainty that might negatively affect the system's performance, if overlooked. These uncertainties may be of aleatory or epistemic types. Aleatory uncertainty includes phenomena such as uncertain operating conditions or imperfect manufacturing processes, while epistemic uncertainty

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stems from the lack of knowledge about a system, such as simplifications in model development. Accounting for these uncertainties transforms the deterministic co-design problem into a stochastic one, requiring the use of appropriate stochastic optimization approaches. A full treatment of uncertainties in co-design formulations requires a characterization of the random processes associated with the stochastic state trajectory decision variables and stochastic control trajectory decision variables. However, in this work, we mainly focus on stochastic co-design formulations when the uncertainty is propagated into the system only from plant design decision variables and problem parameters.

Generally, design approaches under uncertainty can be categorized into robust design optimization (RDO) and reliability-based design optimization (RBDO). While RDO focuses on improving the quality of a product by reducing the sensitivity of the objective function to variations in design, RBDO attempts to maintain design feasibility under uncertainty [11]. In our previous study, we initiated research on the co-design of stochastic dynamic systems by formulating a robust multidisciplinary dynamic system design optimization (R-MDSDO) problem [12,13]. This research takes our previous study to the next level, where a more rigorous evaluation of the probabilistic constraints under aleatory uncertainties is performed using the principles of RBDO [14]. In particular, a balanced co-design method known as multidisciplinary dynamic system design optimization (MDSDO) is used as a basis for an RBDO problem formulation and implementation. Using the sequential optimization and reliability assessment (SORA) strategy for its implementation, we develop an algorithm to solve the reliability-based multidisciplinary dynamic system design optimization (RB-MDSDO) problem, such that the optimization and reliability assessment steps are decoupled from each other and run sequentially [15–18]. The single-loop nature of SORA algorithm in the proposed RB-MDSDO formulation, which is illustrated in Fig. 1, is expected to offer computational advantage when compared with the double-loop reliability-based co-design formulation of Ref. [10]. A performance measure approach (PMA)—which is known to be more robust and efficient compared to the reliability index approach (RIA) [19]—is used along with a first-order reliability method (FORM), to formulate the reliability assessment step and find the most probable point (MPP) corresponding to the targeted reliability level of the performance measure function. The physics-based (analysis-type) dynamic equality constraints are satisfied at the vectors of mean-values of random decision variables, as well as their MPPs. The reliability assessment step finds plant design decision variables that would cause the highest violation of the performance measure function. The general overview of this procedure is presented in Fig. 1, where the mean values of design decision variables, along with the vector of deterministic optimal control trajectory decision variables are passed from the deterministic co-design optimization block into the reliability assessment block, which in turn outputs the MPPs corresponding to the targeted reliability level. The effectiveness of the proposed formulation is then assessed through two case studies, each of which compares the stochastic system solution to the deterministic system solution.

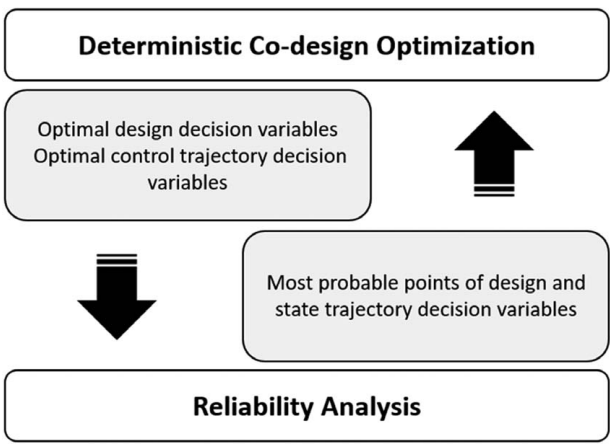


Fig. 1 Overview of the RB-MDSDO procedure

The remainder of this paper is organized in the following manner: Sec. 2 briefly describes MDSDO and how to formulate a general co-design problem within this optimization framework. Section 3 introduces the general concepts of an RBDO problem. Section 4 describes the general formulation of SORA with and without equality constraints. Section 5 integrates SORA within MDSDO and develops the RB-MDSDO problem formulation. Section 6 implements the proposed formulation for two case studies, and finally, Sec. 7 discusses conclusions and future work.

2 Co-Design in MDSDO

Simultaneous integration of plant design and control strategy decisions in optimization problem formulations, also known as co-design, has gained momentum over the last two decades [20–24]. A system-level optimal solution can be obtained through a simultaneous or a nested strategy. In this study, we employ a generalized, integrated dynamic system optimization method known as multidisciplinary dynamic system design optimization or MDSDO [22] to solve co-design problems. MDSDO uses techniques from multidisciplinary design optimization (MDO) and optimal control theory to create a balanced formulation for co-design of dynamic systems. MDSDO enables the definition of a single system-wide objective that is inclusive of plant design and control strategy decisions and can account for a bi-directional coupling between these two disciplines [22]. In its most common form, a co-design problem can be solved within the MDSDO framework through a simultaneous approach.

The nominal simultaneous problem formulation includes the vectors of design decision variables $\mathbf{d}$, state trajectory decision variables $\mathbf{x}(t)$, and control trajectory decision variables $\mathbf{u}(t)$. The objective function ϕ is minimized subject to the vector of inequality constraints $\mathbf{g}(\cdot)$, algebraic equality constraints $\mathbf{h}(\cdot)$, and dynamic system equality constraints $\dot{\mathbf{x}}(t)-\mathbf{f}(\mathbf{d}, \mathbf{x}(t), \mathbf{u}(t), t) = \mathbf{0}$. This formulation can be written as

$$\begin{aligned} \min_{\mathbf{d},\, \mathbf{x}(t),\, \mathbf{u}(t)} \quad & \phi(\mathbf{d}, \mathbf{x}(t), \mathbf{u}(t), t) \\ \text{subject to} \quad & \\ & \mathbf{g}(\mathbf{d}, \mathbf{x}(t), \mathbf{u}(t), t) \le \mathbf{0} \\ & \mathbf{h}(\mathbf{d}, \mathbf{x}(t), \mathbf{u}(t), t) = \mathbf{0} \\ & \dot{\mathbf{x}}(t)-\mathbf{f}(\mathbf{d}, \mathbf{x}(t), \mathbf{u}(t), t) = \mathbf{0} \end{aligned} \tag{1}$$

The above formulation generally requires the application of numerical methods such as direct transcription (DT) to transcribe the infinite-dimensional problem into a finite-dimensional nonlinear programming (NLP) formulation [22,23]. Using DT, a full-time discretization of the states and controls, as well as all functions that are dependent on these variables is performed. Although DT optimization problems have much higher dimension when compared with the other direct methods, they offer high efficiency in dealing with numerically unstable problems [25], exhibit high-order convergence in some cases [25,26], and have a sparse diagonal structure—all of which encourage their application [22,27]. The transcribed version of Eq. (1) can be written as

$$\begin{aligned} \min_{\mathbf{d},\, \mathbf{X},\, \mathbf{U}} \quad & \phi(\mathbf{d}, \mathbf{X}, \mathbf{U}) \\ \text{subject to} \quad & \\ & \mathbf{g}(\mathbf{d}, \mathbf{X}, \mathbf{U}) \le \mathbf{0} \\ & \mathbf{h}(\mathbf{d}, \mathbf{X}, \mathbf{U}) = \mathbf{0} \\ & \boldsymbol{\zeta}(\mathbf{d}, \mathbf{X}, \mathbf{U}) = \mathbf{0} \end{aligned} \tag{2}$$

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where **X** is the matrix of discretized state trajectory decision variables, **U** is the matrix of discretized control trajectory decision variables, and $\boldsymbol{\zeta}$ is the vector of discretized dynamic constraints.

3 Reliability-Based Design Optimization

RBDO attempts to meet specific probabilistic targets for the objective and constraints [28]. To illustrate this further, we introduce a simple optimization problem:

$$\begin{aligned} \min_{\mathbf{s}} \quad & J(\mathbf{s}) \\ \text{subject to} \quad & g_i(\mathbf{s}) \le 0 \quad i = 1, \ldots, n_g \\ & h_j(\mathbf{s}) = 0 \quad j = 1, \ldots, n_h \end{aligned} \tag{3}$$

where J is the objective function and $\mathbf{s} \in R^{n_s}$ is the vector of system design variables with n_s being the number of design variables. Inequality constraints are denoted by $g_i(\mathbf{s})$ and equality constraints are denoted by $h_j(\mathbf{s})$, respectively. Similarly, n_g and n_h refer to the number of inequality and equality constraints, respectively.

A general RBDO problem formulation for a simple optimization problem similar to Eq. (3) but without equality constraints includes the deterministic vector of design variables $\boldsymbol{\mu}_S \in R^{n_r}$ and the random vector $\mathbf{S} \in R^{n_r}$. The probabilistic constraint is described by the performance function $g_i(\mathbf{S})$. RBDO can be implemented through both the traditional double-loop most probable point (MPP) approach or the single-loop MPP approach. Due to the high-computational costs anticipated for the double-loop MPP approach, we implement the proposed reliability-based co-design formulation in a single-loop algorithm. Specifically, the sequential optimization and reliability assessment (SORA) method is employed to implement the RB-MDSDO problem formulation. The general RBDO problem without equality constraints can be described as Ref. [29]

$$\begin{aligned} \min_{\boldsymbol{\mu}_S} \quad & J(\boldsymbol{\mu}_S) \\ \text{subject to} \quad & Pr\{g_i(\mathbf{S}) \le 0\} \ge \alpha_i, \quad i = 1, \ldots, n_g \end{aligned} \tag{4}$$

where $Pr\{\cdot\}$ denotes the probability of success and α_i is the targeted reliability of the ith probabilistic inequality constraint. According to this formulation, reliability is defined as the probability that the vector of random design variables **S** is within the safe region (Fig. 2). This definition is equivalent to the probability that the performance function $g_i(\mathbf{S})$ is less than zero, i.e. $Pr\{g_i(\mathbf{S}) \le 0\}$. Next, we can characterize the reliability of the performance function

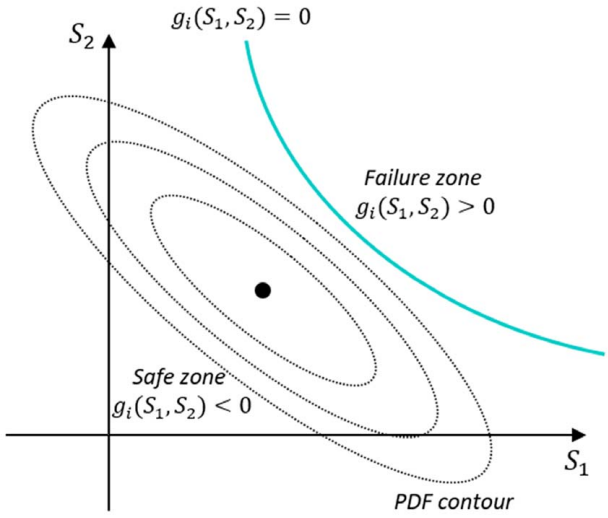


Fig. 2 Performance measure function in the original uncertain space

$g_i(\mathbf{S})$ using the cumulative distribution function $F_{g_i}(0)$ of the ith probabilistic inequality constraint as Ref. [29]

$$Pr\{g_i(\mathbf{S}) \le 0\} = F_{g_i}(0) \ge \alpha_i \tag{5}$$

where this probability can be evaluated using the joint probability density function $f_{\mathbf{S}}(\mathbf{S})$ of all the random variables

$$F_{g_i}(0) = \int_{g_i(\mathbf{S}) \le 0} \cdots \int f_{\mathbf{S}}(\mathbf{S})\, dS_1 \cdots dS_{n_r}, \; i = 1, \ldots, n_g \tag{6}$$

The analytical solution of this equation is very difficult to obtain. Therefore, several approaches have been proposed to evaluate these probabilistic constraints including sampling techniques, local expansion, most probable point, functional expansion, and numerical integration methods [30]. Although sampling techniques such as Monte Carlo simulation (MCS) are easy to implement and able to obtain an accurate solution, their computational time becomes a significant challenge as the number of random variables increases, making them impractical for many engineering applications [31]. The local expansion methods employ a first-order second moment (FOSM) to approximate the probabilistic constraint at the design point $\boldsymbol{\mu}_S$; however, their solution can be highly inaccurate. Numerical integration methods [32] are generally based on the idea that the first few moments of a random variable adequately describe the complete probability density function of that variable. These methods, along with functional expansion methods [33] are more useful for highly nonlinear problems, however, they suffer from the "curse of dimensionality." The MPP approach, on the other hand, can approximate the performance function $g_i(\mathbf{S})$ through a first-order or second-order Taylor series expansion at the most probable point (MPP). While these approaches require the solution of an additional optimization subproblem to find the MPP, they are often faster than sampling, numerical integration, and functional expansion methods and are more accurate than local expansion techniques. Most importantly, the MPP approach is the only known way to address analysis-type equality constraints according to the literature. Therefore, in this study, we employ a MPP approach to estimate Eq. (6).

The optimization subproblem in the MPP formulation of RBDO can be formulated using a reliability index approach (RIA) and performance measure approach (PMA). In RIA, the probabilistic constraint is written in terms of the reliability index β [34], while in PMA, the probabilistic constraint is replaced by the performance measure function $g_i(\mathbf{S})$ [35,36]. These methods are investigated in the literature, and it is shown that the PMA is inherently robust and more efficient in evaluating inactive probabilistic constraints compared to RIA [34,36,37]. The PMA is further developed to address the issues of instability and slow convergence, resulting in improved methods such as advanced mean value (AMV) [38,39], conjugate mean value (CMV) [34,35] and hybrid mean value (HMV) [35,40].

MPP approaches can approximate the performance function $g_i(\mathbf{S})$ through a first-order reliability method (FORM) [19,41,42] or second-order reliability method and (SORM) [42–44]. Generally, FORM predicts the probability of failure with an accuracy that is sufficient for design applications. The general MPP formulation in RBDO approach executes two optimization problems: design optimization in the original random space and reliability analysis in a standard normal space. This requires a transformation from the original random design space **S** to the space of standard normal random space $\mathbf{s}_{u,i}$, using the Rosenblatt transformation [45]

$$\mathbf{s}_{u,i} = \Phi_{cdf}^{-1}[F_{g_i}(\mathbf{s})] \tag{7}$$

where Φ_{cdf}^{-1} is the inverse CDF of standard normal distribution. The point on the target reliability surface $\mathbf{s}_{u,i}^{*}(\beta = \beta_{t_i})$ that maximizes the performance function $g_i(\mathbf{s}_{u,i})$ is called the MPP (Fig. 3). To solve the RBDO problem in this paper, we employ a decoupling approach known as SORA.

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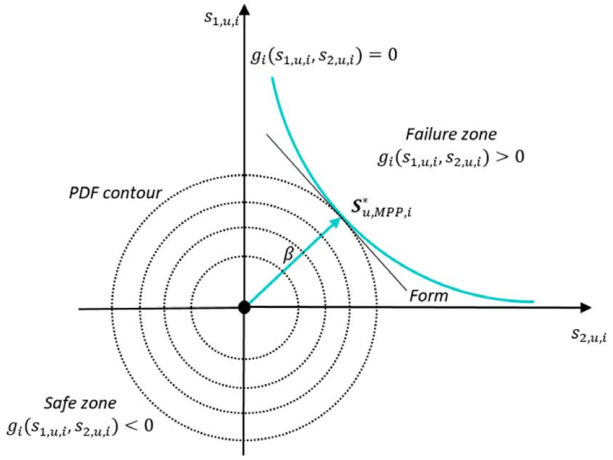


Fig. 3 Performance measure function in the standard normal space

4 Sequential Optimization and Reliability Assessment

The idea behind SORA is to develop a single-loop MPP strategy where the deterministic optimization and the reliability assessment are decoupled from each other and are executed sequentially, until some convergence criteria are achieved. Du and Chen [15,46] employed a series of sequential cycles of optimization and reliability assessment using an efficient MPP search algorithm. In SORA, the deterministic optimization is performed first to achieve an optimal solution, then the reliability analysis is executed to verify the satisfaction of reliability constraints and to provide improvement directions for the next cycle of the SORA algorithm [17]. Here, for the sake of brevity, we only describe the problem formulations in each major step of this algorithm. Further details about the convergence criteria and the flowchart of the algorithm can be found in Refs. [15,46].

4.1 SORA Without Equality Constraints. The SORA algorithm starts with a deterministic problem that includes the mean value vector of the random design variables $\boldsymbol{\mu}_S$. After the first cycle, the deterministic optimization problem uses the information from the reliability assessment step of the previous cycle. In other words, after the first cycle of SORA, the constraints within the deterministic optimization are modified to shift the MPPs into the feasible region of the deterministic bounds. Therefore, we employ shifting vectors, similar to the ones introduced by Du and Chen [18] to establish the equivalent deterministic optimization problem in each cycle of SORA. These shifting vectors are described by $\mathbf{v}$:

$$\mathbf{v}_i^{(m)} = \boldsymbol{\mu}_S^{(m-1)} - \mathbf{s}_{MPP,i}^{*,(m-1)} \tag{8}$$

where $\mathbf{s}_{MPP,i}^{*,(m-1)}$ is the MPP of the vector of uncertain design variables from the $(m-1)$th SORA cycle and $\boldsymbol{\mu}_S^{(m-1)}$ is the vector of mean values from the $(m-1)$th cycle. The deterministic optimization problem for the mth cycle of SORA can then be formulated as

$$\begin{aligned} \min_{\boldsymbol{\mu}_S^{(m)}} \quad & J(\boldsymbol{\mu}_S^{(m)}) \\ \text{subject to} \quad & g_i(\boldsymbol{\mu}_S^{(m)} - \mathbf{v}_i^{(m)}) \le 0, \quad i = 1, \ldots, n_g \end{aligned} \tag{9}$$

The output of Eq. (9), denoted by $\boldsymbol{\mu}_S^{(m)}$, will be the input to the reliability assessment step, where a transformation into the standard normal space will be performed. The MPP of the random design variables associated with the ith performance measure is found through the following formulation

$$\begin{aligned} \min_{\mathbf{s}_{u,i}^{(m)}} \quad & -g_i(\mathbf{s}_{u,i}^{(m)}) \\ \text{subject to} \quad & \left\|\mathbf{s}_{u,i}^{(m)}\right\| - \beta_{t_i} = 0 \\ \text{where} \quad & \mathbf{s}_{u,i}^{(m)} = \Phi_{cdf}^{-1}[F_{g_i}(\mathbf{s}^{(m)})] \end{aligned} \tag{10}$$

This sequential algorithm has been implemented for multiple applications within the RBDO literature. For more details, interested readers are referred to Refs. [16,17,47].

4.2 SORA With Equality Constraints. Equality constraints create a major challenge under uncertainty and have not been studied as extensively as inequality constraints. With the exception of a category of constraints mentioned below, strict satisfaction of equality constraints can generally not be guaranteed. Mattson and Messac classified equality constraint in RDO into two categories: (i) those that cannot be satisfied because of the uncertainty in the problem (design-type) and (ii) those that must be satisfied regardless of the underlying uncertainty in the problem (analysis-type) [48]. Generally, the nature of the existing random variables in the equality constraints plays a significant role in determining their category, and consequently, their formulation in the context of RBDO. Authors in Refs. [48,49] presented a comprehensive guideline to distinguish and formulate each type of equality constraint. Rangavajhala et al. [49] suggested that Type (i) equality constraints should be constrained at their mean to ensure that they remain as close as possible to zero and that their standard deviations remain minimum. An alternative approach to handling design-type constraints is to relax them into a pair of inequality constraints [49]. In this study, we assume that Type (i) equality constrains are already included in the vector of inequality constraints $\mathbf{g}(\cdot)$, and thus, all remaining equality constraints $\mathbf{h}(\cdot)$ are of analysis type (Type (ii)). To satisfy the analysis-type equality constraints within RBDO, Du and Huang developed a numerical procedure that includes the equality constraints in the RBDO model [17]. Defining the analysis-type equality constraint as a function of the random design variables $\mathbf{S} \in R^{n_r}$, we have

$$h_j(\mathbf{S}) = 0, \quad j = 1, \ldots, n_h \tag{11}$$

where n_r denotes the number of random design variables and n_h describes the number of analysis-type equality constraints. Assuming $n_r - n_h$ statistically independent random variables, we can partition the random design variables $\mathbf{S}$ into independent random design variables $\mathbf{S}_x$ and dependent random design variables $\mathbf{S}_y$, such that $\mathbf{S} = [\mathbf{S}_x, \mathbf{S}_y]^T$. We can rewrite Eq. (11) in terms of the functional relationships between $\mathbf{S}_x$ and $\mathbf{S}_y$ [17]:

$$\mathbf{S}_y = \mathbf{a}(\mathbf{S}_x) \tag{12}$$

where $\mathbf{a}$ is the functional relationship between $\mathbf{S}_x$ and $\mathbf{S}_y$. Using the FORM approximation within the PMA for RBDO and denoting the MPP of the constraint function g_i as $\mathbf{s}_{x_{MPP,i}}^{*,(m)}$, the algorithm initially becomes

$$\begin{aligned} \min_{\boldsymbol{\mu}_{S_x}^{(m)}} \quad & J(\boldsymbol{\mu}_{S_x}^{(m)}) \\ \text{subject to} \quad & g_i[\mathbf{s}_{x_{MPP,i}}^{*,(m)}, \mathbf{a}(\mathbf{s}_{x_{MPP,i}}^{*,(m)})] \le 0, \quad i = 1, \ldots, n_g \end{aligned} \tag{13}$$

where $\boldsymbol{\mu}_{S_x}^{(m)}$ is the mean vector of the independent random design variables at the (m)th cycle of SORA. Introducing design variables at the mean values of dependent random design variables $\boldsymbol{\mu}_{S_y}^{(m)}$, their MPPs in $(m-1)$th cycle of SORA for independent random variables as $\mathbf{s}_{x_{MPP,i}}^{*,(m-1)}$, and for dependent random variables as $\mathbf{s}_{y_{MPP,i}}^{*,(m-1)}$, such that $\mathbf{s}_{y_{MPP,i}}^{*,(m-1)} = \mathbf{a}(\mathbf{s}_{x_{MPP,i}}^{*,(m-1)})$, the RBDO model within SORA

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can be written as

$$\begin{aligned}
\min_{\boldsymbol{\mu}_{S_x}^{(m)},\,\boldsymbol{\mu}_{S_y}^{(m)},\,\mathbf{s}_{y_{MPP,i}}^{*,(m)}} \quad & f(\boldsymbol{\mu}_{S_x}^{(m)},\,\boldsymbol{\mu}_{S_y}^{(m)}) \\
\text{subject to} \quad & g_i(\boldsymbol{\mu}_S^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{s}_{y_{MPP,i}}^{*,(m)}) \le 0, \quad i = 1,\ldots,n_g \\
& h_j(\boldsymbol{\mu}_{S_x}^{(m)},\,\boldsymbol{\mu}_{S_y}^{(m)}) = 0, \quad j = 1,\ldots,n_h \\
& h_k(\boldsymbol{\mu}_S^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{s}_{y_{MPP,i}}^{*,(m)}) = 0, \quad k = 1,\ldots,n_h \times n_g, \\
& \qquad\qquad\qquad\qquad\qquad\qquad i = 1,\ldots,n_g
\end{aligned} \tag{14}$$

where $\mathbf{v}_i^{(m)}$ denotes the shifting vector associated with the ith inequality constraint, resulted from the previous cycle of SORA. The equality constraints are satisfied at the mean values of all random variables, as well as their MPPs. Note that since every inequality constraint has its own MPP, the total number of equality constraints at the MPP will be $n_h \times n_g$. Moreover, note that $\mathbf{s}_{x_{MPP,i}}^{*,(m-1)}$ comes from the previous cycle of SORA and is included in the deterministic optimization through the shifting vector $\mathbf{v}_i^{(m)}$. The MPPs of the dependent random variables $\mathbf{s}_{y_{MPP,i}}^{*,(m)}$ are obtained during the (m)th cycle of SORA. This problem is equivalent to Eq. (9) and thus it requires a MPP search algorithm. The reliability assessment is then formulated using a FORM approximation within the PMA. Note that the MPP subproblem must satisfy the analysis-type equality constraints, as well as the usual reliability constraint during optimization.

$$\begin{aligned}
\min_{\mathbf{s}_{u,i}^{(m)},\,\mathbf{s}_y^{(m)}} \quad & -g_i(\mathbf{s}_{u,i}^{(m)},\,\mathbf{s}_y^{(m)}) \\
\text{subject to} \quad & \left\|\mathbf{s}_{u,i}^{(m)}\right\| - \beta_{t_i} = 0 \\
& h_j(\mathbf{s}_{u,i}^{(m)},\,\mathbf{s}_y^{(m)}) = 0, \quad j = 1,\ldots,n_h \\
\text{where} \quad & \mathbf{s}_{u,i}^{(m)} = \Phi_{cdf}^{-1}[F_{g_i}(\mathbf{s}_x^{(m)})]
\end{aligned} \tag{15}$$

5 RB-MDSDO Problem Formulation Using SORA

The integrated nature of co-design formulations results in the propagation of uncertainties from random plant design decision variables to the analysis-type, dynamic system equality constraints, thereby affecting the open-loop optimal control trajectories. To quantify the impact of these uncertainties on the system-level co-design solution, we propose the novel approach for the RB-MDSDO formulation under aleatory uncertainties using SORA.

The single-loop MPP formulation in RB-MDSDO executes the deterministic optimization problem first, followed by the reliability assessment.The deterministic optimization problem, which is the first major block in SORA flowchart of Fig. 4, does not explicitly account for the random variations or processes in the design decision variables or state trajectory decision variables; rather it employs shifting vectors $\mathbf{v}_i^{(m)}$ to shift the MPPs of the design decision variables into the feasible region of the deterministic bounds, while making sure that the state dynamics are satisfied at the mean values, as well as the MPPs. It is still assumed that the overall MPP strategy will consist of the PMA in conjunction with the FORM to solve the problem robustly and efficiently.

This problem formulation includes the vector of mean values of the plant design decision variables $\boldsymbol{\mu}_D^{(m)}$, the mean values of state decision variables $\boldsymbol{\mu}_X^{(m)}(t)$, the MPPs of design decision variables $\mathbf{d}_{MPP,i}^{*,(m)}$, the MPPs of state decision variables $\mathbf{x}_{MPP,i}^{*,(m)}(t)$, the shifting vectors $\mathbf{v}_i^{(m)}$, and the vector of deterministic control variables $\mathbf{u}^{(m)}(t)$ at the (m)th cycle of SORA. The objective function ϕ is optimized over the set of decision variables $[\boldsymbol{\mu}_D^{(m)}, \boldsymbol{\mu}_X^{(m)}(t), \mathbf{x}_{MPP,i}^{*,(m)}(t), \mathbf{u}^{(m)}(t)]$ and is subject to the probabilistic inequality constraints $\mathbf{g}(\cdot)$, and analysis-type equality constraints which constitute algebraic equality constraints $\mathbf{h}(\cdot)$, and dynamic system equality constraints. Note that both algebraic equality constraints and dynamic system equality constraints must be satisfied both at the the mean values and the MPPs of design and state decision variables. This formulation can be described as

$$\begin{aligned}
\min_{\boldsymbol{\mu}_D^{(m)},\,\boldsymbol{\mu}_X^{(m)}(t),\,\mathbf{x}_{MPP,i}^{*,(m)}(t),\,\mathbf{u}^{(m)}(t)} \quad & \phi(\boldsymbol{\mu}_D^{(m)},\,\boldsymbol{\mu}_X^{(m)}(t),\,\mathbf{u}^{(m)}(t)) \\
\text{subject to} \quad & g_i(\boldsymbol{\mu}_D^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{x}_{MPP,i}^{*,(m)}(t),\,\mathbf{u}^{(m)}(t)) \le 0, \quad i = 1,\ldots,n_g \\
& \mathbf{h}(\boldsymbol{\mu}_D^{(m)},\,\boldsymbol{\mu}_X^{(m)}(t),\,\mathbf{u}^{(m)}(t)) = \mathbf{0} \\
& \mathbf{h}_i(\boldsymbol{\mu}_D^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{x}_{MPP,i}^{*,(m)}(t),\,\mathbf{u}^{(m)}(t)) = \mathbf{0}, \quad i = 1,\ldots,n_g \\
& \dot{\boldsymbol{\mu}}_X(t) - \mathbf{f}(\boldsymbol{\mu}_D^{(m)},\,\boldsymbol{\mu}_X^{(m)}(t),\,\mathbf{u}^{(m)}(t)) = \mathbf{0} \\
& \dot{\mathbf{x}}_{MPP,i}^{*,(m)}(t) - \mathbf{f}_i(\boldsymbol{\mu}_D^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{x}_{MPP,i}^{*,(m)}(t),\,\mathbf{u}^{(m)}(t)) = \mathbf{0}, \quad i = 1,\ldots,n_g
\end{aligned} \tag{16}$$

where the shifting vector for the (m)th cycle of SORA can be found using $\mathbf{v}_i^{(m)} = \boldsymbol{\mu}_D^{(m-1)} - \mathbf{d}_{MPP,i}^{*,(m-1)}$. The MPP subproblem in the reliability assessment step can then be formulated as

$$\begin{aligned}
\min_{\mathbf{d}_{u,i}^{(m)}(t),\,\mathbf{x}_i^{(m)}(t)} \quad & -g_i(\mathbf{d}_{u,i}^{(m)},\,\mathbf{x}_i^{(m)}(t),\,\mathbf{u}^{*,(m)}(t)) \\
\text{subject to} \quad & \left\|\mathbf{d}_{u,i}^{(m)}\right\| - \beta_{t_i} = 0 \\
& \mathbf{h}(\mathbf{d}_{u,i}^{(m)},\,\mathbf{x}_i^{(m)}(t),\,\mathbf{u}^{*,(m)}(t)) = \mathbf{0} \\
& \dot{\mathbf{x}}(t) - \mathbf{f}(\mathbf{d}_{u,i}^{(m)},\,\mathbf{x}_i^{(m)}(t),\,\mathbf{u}^{*,(m)}(t)) = \mathbf{0} \\
\text{where} \quad & \mathbf{d}_{u,i}^{(m)} = \Phi_{cdf}^{-1}[F_{g_i}(\mathbf{d}^{(m)})]
\end{aligned} \tag{17}$$

This formulation requires the use of numerical methods such as DT.

The transcribed version of Eq. (16) can be described as

$$\begin{aligned}
\min_{\boldsymbol{\mu}_D^{(m)},\,\mathbf{M}_X^{(m)},\,\mathbf{X}_{MPP,i}^{*,(m)},\,\mathbf{U}^{(m)}} \quad & \phi(\boldsymbol{\mu}_D^{(m)},\,\mathbf{M}_X^{(m)},\,\mathbf{U}^{(m)}) \\
\text{subject to} \quad & g_i(\boldsymbol{\mu}_D^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{X}_{MPP,i}^{*,(m)},\,\mathbf{U}^{(m)}) \le 0, \quad i = 1,\ldots,n_g \\
& \mathbf{h}(\boldsymbol{\mu}_D^{(m)},\,\mathbf{M}_X^{(m)},\,\mathbf{U}^{(m)}) = \mathbf{0} \\
& \mathbf{h}_i(\boldsymbol{\mu}_D^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{X}_{MPP,i}^{*,(m)},\,\mathbf{U}^{(m)}) = \mathbf{0}, \quad i = 1,\ldots,n_g \\
& \boldsymbol{\zeta}(\boldsymbol{\mu}_D^{(m)},\,\mathbf{M}_X^{(m)},\,\mathbf{U}^{(m)}) = \mathbf{0} \\
& \boldsymbol{\zeta}_i(\boldsymbol{\mu}_D^{(m)} - \mathbf{v}_i^{(m)},\,\mathbf{X}_{MPP,i}^{*,(m)},\,\mathbf{U}^{(m)}) = \mathbf{0}, \quad i = 1,\ldots,n_g
\end{aligned} \tag{18}$$

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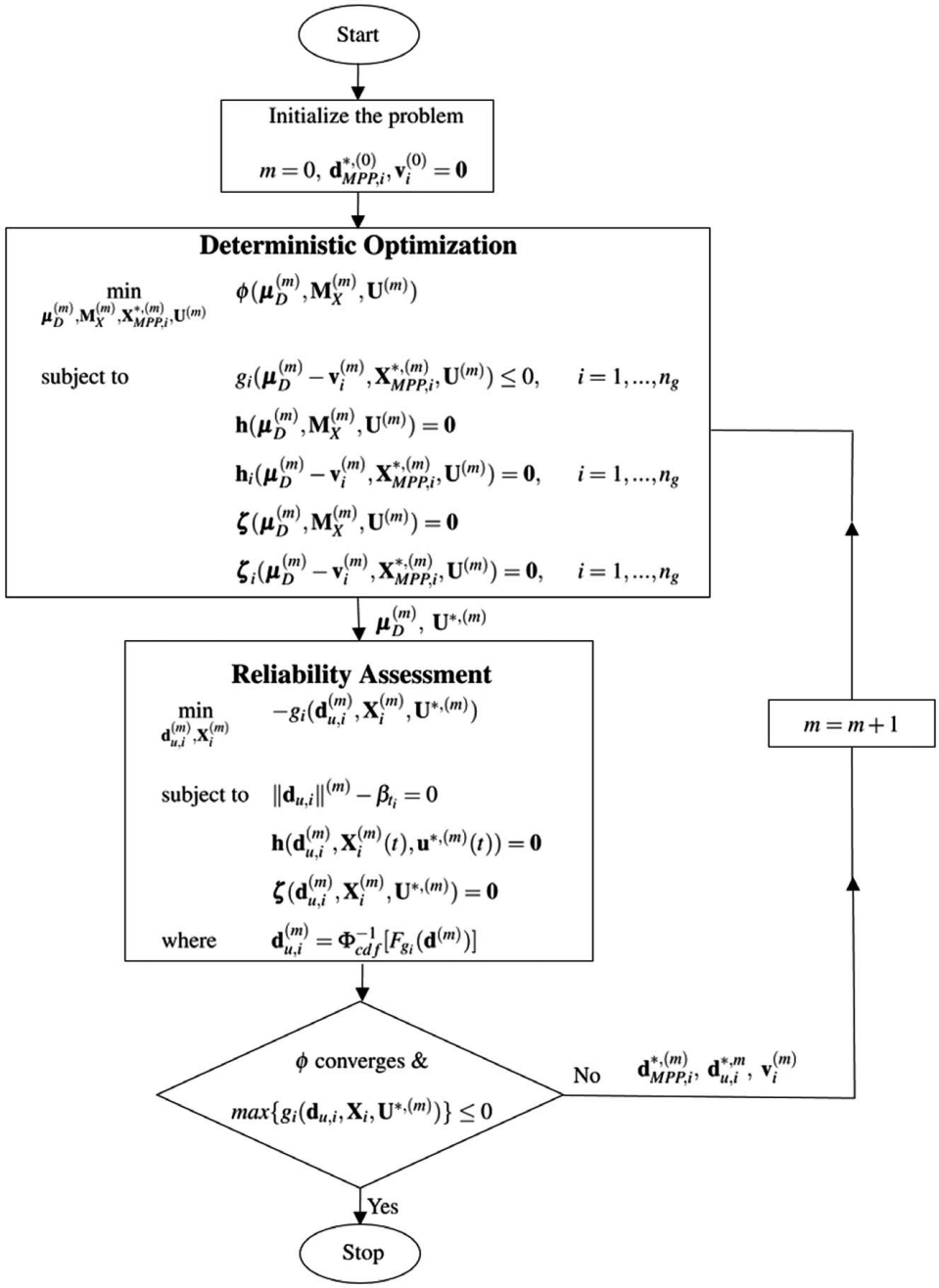


Fig. 4 SORA within RB-MDSDO

where $\mathbf{M}_X^{(m)}$ is the matrix of discretized, state trajectory mean-value decision variables in (m)th SORA cycle. This problem is equivalent to the general RBDO problem and thus requires a reliability assessment algorithm within the standard normal space to find the MPPs. The MPP subproblem can then formulated as

$$\begin{aligned} \min_{\mathbf{d}_{u,i}^{(m)},\, \mathbf{X}_i^{(m)}} \quad & -g_i(\mathbf{d}_{u,i}^{(m)},\, \mathbf{X}_i^{(m)},\, \mathbf{U}^{*,(m)}) \\ \text{subject to} \quad & \left\|\mathbf{d}_{u,i}\right\|^{(m)} - \beta_{t_i} = 0 \\ & \mathbf{h}(\mathbf{d}_{u,i}^{(m)},\, \mathbf{X}_i^{(m)}(t),\, \mathbf{u}^{*,(m)}(t)) = \mathbf{0} \\ & \boldsymbol{\zeta}(\mathbf{d}_{u,i}^{(m)},\, \mathbf{X}_i^{(m)},\, \mathbf{U}^{*,(m)}) = \mathbf{0} \\ \text{where} \quad & \mathbf{d}_{u,i}^{(m)} = \Phi_{cdf}^{-1}[F_{g_i}(\mathbf{d}^{(m)})] \end{aligned} \tag{19}$$

Note that the deterministic optimal control solution from the deterministic optimization block of SORA is a fixed input trajectory into the reliability assessment step. The SORA flowchart of the RB-MDSDO formulation is described in Fig. 4. We note that the output of the deterministic step at the (m)th cycle of SORA to the reliability assessment block is the optimal mean value of independent design variables, $\boldsymbol{\mu}_D^{(m)}$, and optimal control decision trajectory $\mathbf{U}^{*,(m)}$. For the first SORA cycle, since there's no information available from the reliability assessment step, the MPP of the independent random design variables $\mathbf{d}_{MPP,i}^{*,(0)}$ is prescribed as the mean of the independent random design variables $\boldsymbol{\mu}_D^{(0)}$, such that $v_i^{(0)} = \mathbf{0}$. After the initial cycle of SORA, the reliability analysis step inputs the MPP of the independent random design variables of the previous cycle $\mathbf{d}_{MPP,i}^{*,(m-1)}$ to the deterministic step, through the shifting vector. Note that the inclusion of the analysis-type equality

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constraints in the reliability analysis step is to ensure their feasibility.

A major challenge in the implementation of the proposed RB-MDSDO algorithm arises when state trajectory decision variables are bounded through inequality path constraints. As opposed to the conventional static inequality constraints, the dynamic nature of these trajectories necessitates a different approach when the associated performance measure function is formulated. This makes the implementation of the RB-MDSDO problem formulation a challenging task. One could treat each discretized path constraint at every single collocation point as an inequality constraint—employed within RB-MDSDO problem formulation as a performance measure function. This, however, is not practical as the number of performance measure functions will be directly influenced by the total number of collocation points in the discretized problem. To address this issue, we identify the least reliable design point for each probabilistic path constraint by maximizing the integral of the probabilistic path constraint at every reliability assessment step. This approach enables the determination of plant designs and state trajectories that would cause the highest constraint violation over the whole trajectory under uncertainty. The stopping criteria is then set as the convergence of the optimization subproblems, as well as the satisfaction of the performance measure function at the MPP. It is noteworthy to mention that while this approach is limiting in the sense that it does not provide insights into the characteristics of this random process, it does address the reliability requirements of the RB-MDSDO formulation when the uncertainty is propagated into the system only from plant design decision variables or problem parameters.

6 RB-MDSDO Implementations

Implementation of the proposed formulation is investigated in two case studies. The first example is a simple analytical constrained Van der Pol oscillator optimal control problem, modified for a co-design problem formulation. The original problem was taken from Refs. [50,51]. While extremely simplified, this problem offers obvious advantages in identifying implementation challenges. The second example is the range maximization co-design problem of a hang glider with physical design variables, intended to illustrate more complex dynamic problems. We solved both the deterministic and the reliability assessment steps of the RB-MDSDO SORA algorithm on a 32 GB of RAM and an Intel(R) Xenon(R) CPU E5-2637 v3 @ 3.50 GHz processor system.

6.1 Van Der Pol Oscillator. The deterministic problem is to find the optimal plant design $\mathbf{d} = [d_1, d_2]$, state trajectories $\mathbf{x}(t) = [x_1(t), x_2(t)]$, and control trajectory $u(t)$ to minimize the objective function ϕ:

$$\min_{\mathbf{d},\,\mathbf{x}(t),\,u(t)} \quad \phi = \int_0^5 \left[x_1^2(t) + x_2^2(t) + u^2(t)\right] \mathrm{d}t$$

subject to

$$\begin{aligned}
&\dot{x_1}(t) - x_1(t)(1 - x_2^2(t)) + D_2 D_1 x_2(t) - u(t) = 0\\
&\dot{x_2}(t) - x_1(t) = 0\\
&\mathbf{d}_{\min} \le \mathbf{d} \le \mathbf{d}_{\max}\\
&\mathbf{x}_{\min} \le \mathbf{x} \le \mathbf{x}_{\max}\\
&u_{\min} \le u(t) \le u_{\max}\\
&g = -0.4 - x_1(t) \le 0
\end{aligned} \tag{20}$$

where $\mathbf{d}_{\min} = [-5, -5]$, $\mathbf{d}_{\max} = [5, 5]$, $\mathbf{x}_{\min} = [-2, -2]$, and $\mathbf{x}_{\max} = [2, 2]$. The lower and upper bounds on $u(t)$ are −0.5 and 1.5, respectively. The problem is subject to the following boundary conditions: $\mathbf{x}(0) = [0, 1]^T$. Assuming that both design variables D_1 and D_2 are uncertain normal variables with $\sigma_{D1} = 0.02$ and $\sigma_{D2} = 0.03$, and the reliability target of $\beta_t = 3$ for the performance measure function $g: -0.4 - x_1(t) \le 0$, the objective for the deterministic step of SORA can be formulated as

$$\min_{\boldsymbol{\mu}_D^{(m)},\,\boldsymbol{\mu}_X^{(m)}(t),\,\mathbf{x}_{MPP}^{*,(m)}(t),\,u^{(m)}(t)} \phi = \int_0^5 \left[\mu_{X_1}^{2,(m)}(t) + \mu_{X_2}^{2,(m)}(t) + u^{2,(m)}(t)\right] \mathrm{d}t \tag{21}$$

which is subject to the following constraints:

$$\begin{aligned}
&\dot{\mu}_{X_1}^{(m)}(t) - \mu_{X_1}^{(m)}(t)(1 - \mu_{X_2}^{2,(m)}(t)) + \mu_{D_2}^{(m)} \mu_{D_1}^{(m)} \mu_{X_2}^{(m)}(t) - u^{(m)}(t) = 0\\
&\dot{\mu}_{X_2}^{(m)}(t) - \mu_{X_1}^{(m)}(t) = 0\\
&\dot{x}_{1,MPP}^{*,(m)}(t) - x_{1,MPP}^{*,(m)}(t)(1 - x_{2,MPP}^{2,*,(m)}(t)) + d_{2,MPP}^{*,(m-1)} d_{1,MPP}^{*,(m-1)} x_{2,MPP}^{*,(m)}(t) - u^{(m)}(t) = 0\\
&\dot{x}_{2,MPP}^{*,(m)}(t) - x_{1,MPP}^{*,(m)}(t) = 0\\
&\boldsymbol{\mu}_{D_{\min}} + 3\boldsymbol{\sigma}_d \le \boldsymbol{\mu}_D \le \boldsymbol{\mu}_{D_{\max}} - 3\boldsymbol{\sigma}_d\\
&\boldsymbol{\mu}_{X_{\min}} \le \boldsymbol{\mu}_X(t) \le \boldsymbol{\mu}_{X_{\max}}\\
&u_{\min} \le u(t) \le u_{\max}\\
&G = -0.4 - X_1^*(t) \le 0
\end{aligned} \tag{22}$$

Note that the design space of the plant decision variables is reduced under uncertainty. The reliability assessment step then pertains to finding the MPP associated with the maximum violation of the performance measure trajectory within the standard normal space $\mathbf{d}_{u,i}$. Due to the assumption of normality for the uncertain plant design variables, the Rosenblatt transformation [45] would be linear:

$$\mathbf{d}_{u,i} = \frac{\mathbf{D} - \boldsymbol{\mu}_D}{\boldsymbol{\sigma}_D} \tag{23}$$

Then, the reliability assessment step can be formulated as

$$\min_{\mathbf{d}_u^{(m)},\,\mathbf{x}^{(m)}(t)} \quad -\int_0^5 \left[-0.4 - x_1^{(m)}(t)\right] \mathrm{d}t$$

subject to

$$\begin{aligned}
&\left\|\mathbf{d}_u^{(m)}\right\| - \beta_{t_i} = 0\\
&\dot{x}_1^{(m)}(t) - x_1^{(m)}(t)(1 - x_2^{2,(m)}(t)) + (\mu_{D_2}^{(m)} + d_{2,u}\sigma_2)(\mu_{D_1}^{(m)}\\
&\quad + d_{1,u}\sigma_1) x_2^{(m)}(t) - u^{*,(m)}(t) = 0\\
&\dot{x}_2^{(m)}(t) - x_1^{(m)}(t) = 0
\end{aligned} \tag{24}$$

The problem was solved using the commercially available MATLAB©-based software, GPOPS-II [52], which approximates the continuous-time optimal control problems using a new class of variable-order Gaussian quadrature methods. GPOPS-II employs a DT-based approach to transcribe the infinite dimensional problem into a problem formulation similar to that of Eq. (2). The transcribed problem is then solved with IPOPT [53]. The key GPOPS-II settings [52] for this example are introduced in Table 1.

Table 2 compares the solution of the deterministic and the proposed RB-MDSDO problem formulations. The results indicate the critical impact of uncertainty on the optimal solution of co-design formulations within MDSDO and the necessity to account for this uncertainty through stochastic approaches such as the proposed RB-MDSDO formulation. The solution converged in $m = 7$ SORA cycles, and the vector of plant design variables

Table 1 GPOPS-II settings

Field	Setting
Mesh method	hp-LiuRao-Legendre
Mesh tolerance	10^{-6}
NLP solver	IPOPT
IPOPT tolerance	10^{-7}
IPOPT max iterations	5000

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Table 2 Co-design results for Van der Pol oscillator

	d_1	d_2	$d^*_{1,MPP}$	$d^*_{2,MPP}$	J	t (s)
MDSDO[a]	0.324	0.21	–	–	1.969	0.82
RB-MDSDO[b]	0.008	0.014	−0.035	−0.051	1.978	8.02
Diff[c]	96.31%				0.46%	

[a]Deterministic MDSDO solution.
[b]RB-MDSDO solution
[c]Percent difference

has changed significantly compared to the deterministic case (96.31%). Although the value of the objective function has increased by 0.46% in the RB-MDSDO approach, the associated solution is more reliable compared to the deterministic case. The probability of failure for the probabilistic path constraint, G, was benchmarked through MCS, where $n = 100{,}000$ points were sampled about the MDSDO and RB-MDSDO solutions, respectively. The simulation results show that the probability of failure has dropped from $Pr_{f,MDSDO}\{G\} = 0.294$ in MDSDO to $Pr_{f,RB-MDSDO}\{G\} = 0.03097$ in RB-MDSDO—indicating the effectiveness of the proposed approach. The probability of failure can be further improved by increasing the reliability target or using SORM. Figures 5 and 6 show the results of the MCS for the state decision trajectories about the solution of the MDSDO and RB-MDSDO approach, respectively. A sample size of $n = 1,000$ is used to create the MCS plots. From this figure, it is evident that the resultant trajectories violate G more often than the RB-MDSDO solution shown in Fig. 6. The optimal control trajectory for the RB-MDSDO solution is also illustrated in Fig. 7.

6.2 Hang Glider. This problem is a modified version of the range maximization flight of a hang glider. Originally taken from Ref. [54], the problem is modified to include design variables. Here, the hang glider is approximately described as a point mass subject to its weight W, a lift force L and drag force D_r. The motion of the glider is described by four state equations. A distribution for the thermal updraft with respect to the horizontal distance x_1 is given by the upward wind velocity [55]:

$$v_a(x_1) = v_{a_{\max}} \exp\left(-\left(\frac{x_1}{R}-2.5\right)^2\right)\left(1-\left(\frac{x_1}{R}-2.5\right)^2\right) \tag{25}$$

where $R = 100$ m and $v_{a_{\max}} = 2.5$ m/s. The difference between initial and final altitude is used to maximize the horizontal range. The problem is modified to include the calculation of a trapezoidal wing surface area, such that a_1 and a_2 are the bases and b is the height of the wing. The problem is constructed with a plant design $\mathbf{d} = [a_1, a_2, b]^T$, state trajectories $\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t), x_4(t)]^T$ and $u(t) = c_l(t)$. The state trajectories $x_1(t)$ and $x_2(t)$ are horizontal and vertical positions, respectively, and $x_3(t)$ and $x_4(t)$ denote horizontal and vertical absolute velocity components of the glider. The glider is controlled through the lift coefficient $c_l(t)$. The deterministic co-design problem using MDSDO can be formulated as

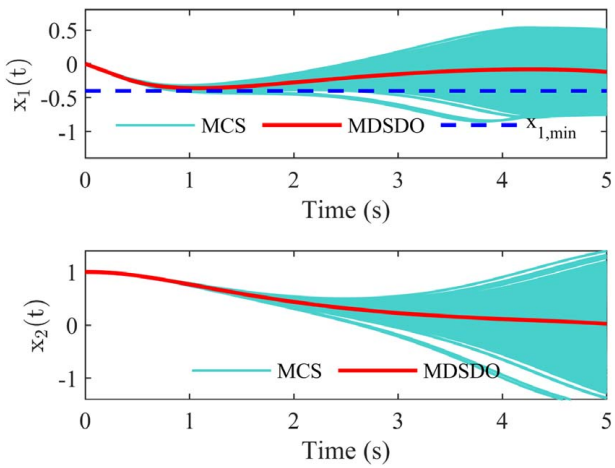


Fig. 5 Van der Pol oscillator: MCS about the solution of MDSDO, $P_f = 0.294$

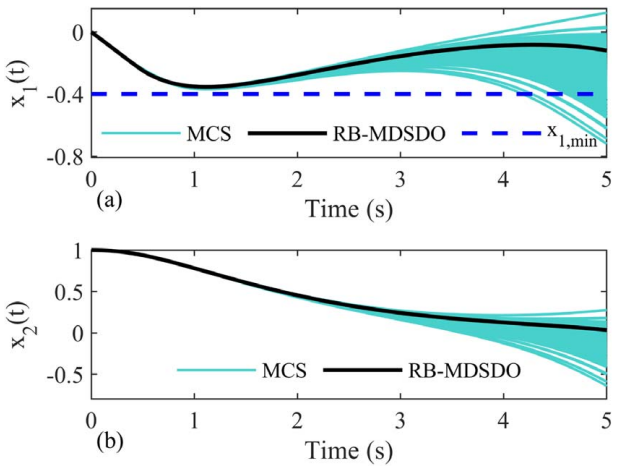


Fig. 6 Van der Pol oscillator: MCS about the solution of RB-MDSDO for state trajectories: (*a*) $x_1(t)$ and (*b*) $x_2(t)$

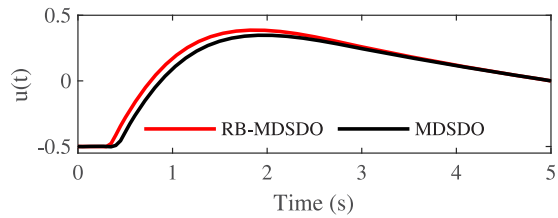


Fig. 7 Van der Pol oscillator: optimal control trajectory $u(t)$

$$\min_{\mathbf{d}(t),\,\mathbf{x}(t),\,\mathbf{u}(t),\,x_1(t_f),\,t_f} \quad -x_1(t_f) \tag{26a}$$

subject to

$$\dot{x}_1(t) - x_3(t) = 0 \tag{26b}$$

$$\dot{x}_2(t) - x_4(t) = 0 \tag{26c}$$

$$\dot{x}_3(t) - \frac{1}{m}(-L\sin(\eta) - D_r\cos(\eta)) = 0 \tag{26d}$$

$$\dot{x}_4(t) - \frac{1}{m}(L\cos(\eta) - D_r\sin(\eta) - W) = 0 \tag{26e}$$

$$L/D_r - 11 \le 0 \tag{26f}$$

$$\frac{(2b + f_l)^2}{S} - 10 \le 0 \tag{26g}$$

$$S - S_{\max} \le 0 \tag{26h}$$

$$x_{\min} \le \mathbf{x}(t) \le x_{\max} \tag{26i}$$

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$$a_{1\min} \le a_1 \le a_{1\max} \tag{26j}$$

$$a_{2\min} \le a_2 \le a_{2\max} \tag{26k}$$

$$b_{\min} \le b \le b_{\max} \tag{26l}$$

where $a_{1\min}=0.5$, $a_{1\max}=3.5$, $a_{2\min}=1.5$, $a_{2\max}=4$, $b_{\min}=4$, and $b_{\max}=10$. The lower and upper bounds on state trajectory decision variables are $x_{\min}=[0,\ 850,\ 0,-10]$ and $x_{\max}=[4000,\ 1100,\ 20,\ 10]$, respectively, and $S_{\max}=19$. Total mass of the pilot and glider is $m=100$ kg and $W=mg$ is the corresponding weight with g being the gravitational constant. f_l is the fuselage length and assumed to be constant. Equations (26*b*)–(26*d*) are dynamic equations and Eqs. (26*e*)–(26*g*) represent the bounds on glide ratio, aspect ratio, and the wing surface area and are included in the design to keep the problem realistic [56]. The relative velocity v_r is at angle η relative to the horizontal plane and calculated by

$$\begin{aligned} v_r &= \sqrt{x_3^2 + (x_4 - v_a(x_1))^2} \\ \eta &= \arctan\left(\frac{x_4 - v_a(x_1)}{x_3}\right) \end{aligned} \tag{27}$$

Lift L and drag D_r forces are calculated through:

$$\begin{aligned} L &= 0.5 c_l \rho S {v_r}^2 \\ D_r &= 0.5 c_d \rho S {v_r}^2 \\ c_d &= 0.034 + 0.069662 {c_l}^2 \end{aligned} \tag{28}$$

where c_l and c_d are lift and drag coefficients and S describes the area of trapezoidal wings, calculated through $S=2[1/2(a_1+a_2)b]$. Air density corresponding to standard pressure and temperature at a height of 1000 m is equal to $\rho=1.3$ kg/m^3. The glider is controlled through the lift coefficient and is restricted by

$$c_l \le c_{l\max} = 1.4 \tag{29}$$

The problem is solved with the following boundary conditions:

$$\begin{aligned} &x_1(t_0) = 0\,\text{m} \\ &x_2(t_0) = 1000\,\text{m} \\ &x_2(t_f) = 900\,\text{m} \\ &x_3(t_0) = x_3(t_f) = 13.23\,\text{m/s} \\ &x_4(t_0) = x_4(t_f) = -1.288\,\text{m/s} \end{aligned} \tag{30}$$

In order to formulate the RB-MDSDO problem, we assume a reliability target of $\beta=3$. The standard deviation of the independent plant design variables is assumed to be $\sigma_{a_1}=0.1$, $\sigma_{a_2}=0.1$, $\sigma_b=0.1$. We note that since Eq. (26*f*) is completely deterministic (when simplified, it's only a function of c_l, which is a deterministic variable), we are left with two performance measure functions, namely Eqs. (26*g*) and (26*h*). These performance measure functions are distinguished by index $i=1$ and $i=2$, respectively. Using a single-loop MPP formulation in RB-MDSDO, along with FORM and PMA, the objective for the deterministic step of SORA for hang glider co-design can be formulated as

$$\min_{\boldsymbol{\mu}_d^{(m)},\, \boldsymbol{\mu}_x^{(m)}(t),\, \mathbf{x}_{MPP,i}^{*,(m)}(t),\, \mathbf{u}^{(m)}(t),\, \boldsymbol{\mu}_{x_1}^{(m)}(t_f),\, t_f} \quad -\mu_{x_1}^{(m)}(t_f) \tag{31}$$

where (m) denotes the mth cycle of SORA. This is subject to the following constraints:

$$\begin{aligned} &\dot{\mu}_{x_1}^{(m)}(t) - \dot{\mu}_{x_3}^{(m)}(t) = 0 \\ &\dot{\mu}_{x_2}^{(m)}(t) - \dot{\mu}_{x_4}^{(m)}(t) = 0 \\ &\dot{\mu}_{x_3}^{(m)}(t) - \frac{1}{m}(-\mu_L \sin(\mu_\eta) - \mu_{D_r}\cos(\mu_\eta)) = 0 \\ &\dot{\mu}_{x_4}^{(m)}(t) - \frac{1}{m}(\mu_L \cos(\mu_\eta) - \mu_{D_r}\sin(\mu_\eta) - W) = 0 \\ &\dot{x}_{1_{MPP,i}}^{(m)}(t) - \dot{x}_{3_{MPP,i}}^{(m)}(t) = 0 \quad i = 1, 2 \\ &\dot{x}_{2_{MPP,i}}^{(m)}(t) - \dot{x}_{4_{MPP,i}}^{(m)}(t) = 0 \quad i = 1, 2 \\ &\dot{x}_{3_{MPP,i}}^{(m)}(t) - \frac{1}{m}(-L_{MPP,i}\sin(\eta_{\text{MPP,i}}) - \\ &D_{r_{MPP,i}}\cos(\eta_{\text{MPP,i}})) = 0 \quad \text{i} = 1, 2 \\ &\dot{x}_{4_{MPP,i}}^{(m)}(t) - \frac{1}{m}(L_{MPP,i}\cos(\eta_{\text{MPP,i}}) - \\ &D_{r_{MPP,i}}\sin(\eta_{\text{MPP,i}}) - \text{W}) = 0 \quad \text{i} = 1, 2 \\ &\frac{L}{D_r} - 11 \le 0 \\ &G_1 = \frac{(2b_{MPP,1} + f_l)^2}{S_{MPP,1}} - g_{1,\max} \le 0 \\ &G_2 = S_{MPP,2} - S_{\max} \le 0 \\ &\boldsymbol{\mu}_{x_{\min}} \le \boldsymbol{\mu}_x(t) \le \boldsymbol{\mu}_{x_{\max}} \\ &\mu_{a_{1_{\min}}} + 3\sigma_{a_1} \le \mu_{a_1} \le \mu_{a_{1_{\max}}} - 3\sigma_{a_1} \\ &\mu_{a_{2_{\max}}} + 3\sigma_{a_2} \le \mu_{a_2} \le \mu_{a_{2_{\max}}} - 3\sigma_{a_2} \\ &\mu_{b_{\min}} + 3\sigma_b \le \mu_b \le \mu_{b_{\max}} - 3\sigma_b \end{aligned} \tag{32}$$

Note that in this formulation, the dynamic system equality constraints are satisfied at both the vector of mean values, as well as the MPPs. All the deterministic decision variables including the vector of mean values of plant design variables $\boldsymbol{\mu}_d^{(m)}$, optimal control trajectory $\mathbf{u}^{(m)}(t)$, and $t_f^{(m)}$ will be inputs to the reliability assessment step. Moreover, under uncertainty, it is customary to push the design space inward so as to avoid the boundaries of the design space under uncertainty (Table 3). This would be a trivial case of making the box constraints reliable when they have a Gaussian distribution. Solving the MPP subproblem in the reliability assessment step requires relaxation of the final boundary conditions on the system. This makes sense because it is reasonable to assume that the initial conditions of the systems could still be prescribed by the designer under uncertainty; however, this uncertainty will be propagated into the system and thus, we cannot guarantee that the final conditions will always be met under uncertainty. Thus, the MPP subproblem

Table 3 GPOPS-II settings

Field	Setting
Mesh method	hp-LiuRao-Legendre
Mesh tolerance	10^{-3}
NLP solver	IPOPT
IPOPT tolerance	10^{-4}
IPOPT max iterations	5000

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Table 4 Co-design results for hang glider

	a_1	a_2	b	$x_1(t_f)$	t (s)
MDSDO[a]	1.62	2.48	4.63	1,290.4	13.58
RB-MDSDO[b]	1.44	2.46	4.3	1,272.7	289.8
Diff[c]	6.85%			1.37%	

[a]Deterministic MDSDO solution.
[b]RB-MDSDO solution.
[c]Percent difference.

must be solved with the following conditions:

$$\begin{aligned} x_1^{(m)}(t_0) &= 0\,\text{m} \\ x_2^{(m)}(t_0) &= 1000\,\text{m} \\ x_3^{(m)}(t_0) &= 13.23\,\text{m/s} \\ x_4^{(m)}(t_0) &= -1.288\,\text{m/s} \end{aligned} \tag{33}$$

$$\begin{aligned} &\min_{\mathbf{d}_{u,i},\,\mathbf{x}_i(t)} \quad -g_i \\ &\text{subject to} \\ &\dot{x}_{1,i}(t) - x_3(t) = 0 \\ &\dot{x}_{2,i}(t) - x_4(t) = 0 \\ &\dot{x}_{3,i}(t) - \frac{1}{m}(-L_i\sin(\eta_i) - D_{ri}\cos(\eta_i)) = 0 \\ &\dot{x}_{4,i}(t) - \frac{1}{m}(L_i\cos(\eta_i) - D_{ri}\sin(\eta_i) - W) = 0 \\ &\|\mathbf{d}_{u,i}\| - \beta_{t_i} = 0 \end{aligned} \tag{34}$$

where

$$\begin{aligned} \mathbf{d}_{u,i} &= \Phi_{cdf}^{-1}[F_{g_i}(\mathbf{d})] \\ g_1 &= \frac{(2b + f_l)^2}{S} - g_{1,max} \le 0 \\ g_2 &= S - S_{\max} \end{aligned} \tag{35}$$

The problem was solved using the commercially available software, GPOPS-II [52] in $m = 4$; SORA cycles, and the results from both deterministic MDSDO and RB-MDSDO formulations are compared in Table 4. The plant design vector has changed by 6.85% compared to the deterministic case. Also, the maximum range is reduced by 1.37% in the RB-MDSDO approach. Although less favorable, the associated risk with this design is much less. The MPPs associated with plant design variables at $m = 4$ SORA cycle for $i = 1$ and $i = 2$ are found to be $a_{1,MPP,i}^{*,(4)} = 1.62$, $a_{2,MPP,i}^{*,(4)} = 2.63$, and $b_{MPP,i}^{*,(4)} = 4.47$. and $a_{1,MPP,i}^{*,(4)} = 1.25$, $a_{2,MPP,i}^{*,(4)} = 2.27$, and $b_{MPP,i}^{*,(4)} = 4.44$, respectively. The probabilities of failure for both of the performance measure functions in Eq. (35) are then benchmarked MCS with $n = 100,000$ samples as indicated in Table 5.

The optimal control and state trajectories for the RB-MDSDO solution are illustrated in Figs. 8 and 9, respectively. The state trajectories are plotted against state trajectories that are obtained by performing MCS about the solution of RB-MDSDO. A sample size of $n = 1000$ is used to create these figures. It can be observed that under uncertainty, optimal control trajectories and states of the system tend to deviate from the solution of the deterministic problem.

Table 5 Probabilities of failure using Monte Carlo Simulation with $n = 100,000$ samples

	MDSDO	RB-MDSDO
$Pr_f\{G_1\}$	0	0
$Pr_f\{G_2\}$	0.4986	0.0468

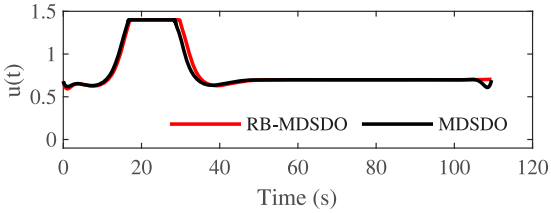


Fig. 8 Hang glider: optimal control trajectory

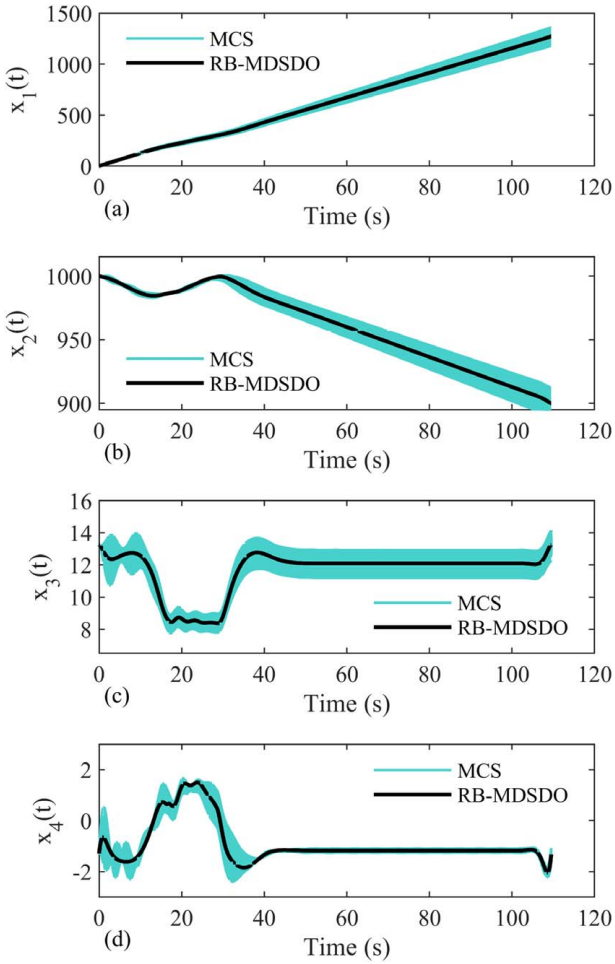


Fig. 9 Hang glider: MCS about the solution of RB-MDSDO for state trajectories

7 Conclusions and Future Work

Addressing co-design for stochastic dynamic systems is important since most real systems have some level of uncertainty, and thus neglecting this behavior could lead to co-design solutions that are either suboptimal or simply infeasible. This study develops and implements the first known reliability-based MDSDO approach for stochastic co-design problems when uncertainties stem from design decision variables and/or problem parameters. The proposed formulation was applied to two case studies and results indicated the

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significant impact of RB-MDSDO approach on the overall system solution, as well as the performance measure.

A unique challenge for the RB-MDSDO implementation is the trajectory-based performance measure functions. In this study, we maximized the integral of the probabilistic path constraint at every reliability assessment step. This study offers a practical approach towards incorporating the uncertainty from design decision variables and problem parameters in co-design problems. However, the uncertainty could be directly imposed on state and/or control trajectory decision variables, requiring full characterization of these random processes. This issue must be investigated in future research. In addition, this work focuses on a single-loop MPP-based formulation for RB-MDSDO. The assumption was that the double-loop MPP-based formulations would incur a high computational expense on the problem and thus, might not be practical for the co-design formulation. Yet, it is noteworthy to mention that despite their double-loop nature, the MPP subproblems of the traditional double-loop MPP formulations are generally smaller problems, as they do not need to satisfy the (expensive) dynamic system constraints. Therefore, it is important to identify whether or not single-loop MPP-based approaches offer any computational advantages over their double-loop MPP counterparts, and if they do, whether there are any significant compromises in solution accuracy. Selecting the most effective MPP implementation for RB-MDSDO is therefore necessary in future research. The application of the proposed approach to a real-world, large-scale engineering problem must also be investigated in future research in order to address scalability concerns and computational expense of the proposed approach. Finally, since any practical implementation of co-design problems requires a closed-loop control structure, an important future direction for this research would be to close the control loop in the co-design problem. For the current study, this structure results in stochastic optimal control trajectory decisions.

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